

MAT 426, Class 20: Singular Value Decomposition

Definition. If A is an $m \times n$ matrix, the **singular values** of A are the _____ roots of the _____ of $A^T A$ and are denoted by $\sigma_1, \dots, \sigma_n$. It is conventional to arrange the singular values so that $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$

Example 1. Find the singular values of $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Example 2. Find the singular values of $A = \begin{bmatrix} 4 & 6 \\ 0 & 4 \end{bmatrix}$.

SINGULAR VALUE DECOMPOSITION *Let A be an $m \times n$ matrix with singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ and $\sigma_{r+1} = \sigma_{r+2} = \cdots = \sigma_n = 0$. Then there exists an $m \times m$ orthogonal matrix U , an $n \times n$ orthogonal matrix V and an $m \times n$ matrix diagonal matrix Σ such that $A = U\Sigma V^T$.*

Proof:

Example 3. Find a singular value decomposition of $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Example 4. Find a singular value decomposition of $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Example 5. Find a singular value decomposition of $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$.